

Advay Kadam

732-986-7792 | advay.kadam10@gmail.com | linkedin.com/in/advay-kadam | github.com/Advay-k1

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Computer Science and Statistics, Minor in Mathematics | GPA: 3.9

December 2026

- **Relevant Courses:** Algorithms and Models of Computation, Machine Learning, Abstract Linear Algebra, Time Series Analysis, Statistical Modeling, Data Structures and Algorithms, Computer Architecture

EXPERIENCE

Bank of New York Mellon

June 2025 – Present

Data Science and Engineering Intern

Jersey City, NJ

- Integrated RAG based agent to automate metadata security classification across **4,000+** datasets in data lake
- Automated dataset tagging via chain of thought reasoning agent, reducing manual audit time by **15 hours/week**
- Reduced end-to-end latency by **83%** for multi-agent pipeline through multithreaded input processing
- Enhanced metadata-driven quality checks with RAG agent, boosting detection of low-quality metadata by **35%**

Procter & Gamble

May 2024 – Aug 2024

Software Engineering Intern

Cincinnati, OH

- Cut sales ETL pipeline refresh time by **60%** on **200k+** records via optimized SQL and PySpark transformations
- Accelerated sales dashboard update speed by **30%**, improving responsiveness for **300+** analysts
- Implemented statistical anomaly detection using Isolation Forests with Python to flag deviations in batch loads
- Fine-tuned YOLOv9 model on retail image data to **95%** accuracy and executed model on Azure ML pipeline

RESEARCH AND CAMPUS INVOLVEMENT

Siebel Center of Computing and Data Science

Jan 2025 – Present

Course Assistant for Data Structures and Algorithms

Urbana, IL

- Led office hours and discussion sessions on abstract data types, C++ algorithms, and complexity analysis
- Developed and maintained algorithmic exercises (e.g., nearest neighbor search) for **400+** enrolled students

Disruption Lab at Gies

Aug 2024 – May 2025

Lead Software Engineer

Urbana, IL

- Led a team of **6** students to train transfer learning model to detect AI-generated deepfake images with TensorFlow
- Trained and deployed model in AWS SageMaker, boosting accuracy to **96%** via hyperparameter tuning

National Center for Supercomputing Applications

Aug 2023 – May 2024

Undergraduate Researcher

Urbana, IL

- Analyzed 2PB+ of real-time network metadata using Pandas, NumPy, Shell scripting to assess PQC resilience
- Identified that over **99%** of SSH traffic lacked quantum-resistant encryption in collaboration with Dr. Phuong Cao
- Developed Flask dashboard to parse Zeek logs and deliver live metrics on PQC algorithm usage and vulnerability

PROJECTS

IMC Prosperity (International Trading Competition) | *Python, Pandas, Matplotlib, Seaborn*

- Placed in the top **3%** in a 15-day global trading challenge, writing Python scripts for algorithmic trading
- Applied Backtesting and implemented strategies related to market making, Black-Scholes pricing, and regression

Regression Library | *C++, Xtensor, Boost, CMake*

- Built C++ library for multiple linear regression and hypothesis testing for reduced regression coefficients
- Implemented stochastic/batch gradient descent algorithms and confidence intervals for true parameter estimation

Time Series Regression Analysis of Sea Surface Temperatures | *R*

- Performed research on creating robust regression models for forecasting sea surface temperatures during El Niño
- Modeled autocorrelated errors and validated residual assumptions via ACF/PACF plots and Ljung-Box tests

Speech Transcription for Academia | *Python, OpenAI API, Hugging Face*

- Curated and preprocessed **150+** hours of academic audio to build custom dataset for OpenAI Whisper fine-tuning
- Improved transcription accuracy by **25%** for non-standard and domain-specific lecture content

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Java, JavaScript, HTML/CSS

Tools: PyTorch, Tensorflow, NumPy, Pandas, Flask, Azure, AWS SageMaker, Apache Spark/PySpark, Matplotlib, Git

Awards | **3rd place** at UIUC Spring 2024 Datathon, **2nd place** at UIUC Fall 2023 Datathon, **2x Dean's List**